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Homework chapter 1

1- the continuum assumption is that $K \ll 1$, so the continuum assumption is not valid when $K \ll \frac{\lambda}{L}$

Knudsen number: $K = \frac{\lambda}{L}$ with λ the mean free path of molecule
L the characteristic length of the physical problem

Ex: Gas flow in microchannels

2. the fluid rotation: the particles rotates around axes
the rotation rate is link to the vorticity ξ

$\xi = \text{curl } \vec{V} = 2\omega$. $\xi = \begin{pmatrix} \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \\ \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \\ \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \end{pmatrix}$

3. Linear deformation: particle's sides stretch or contract
dilation rate $= \nabla \cdot \vec{V}$

4. the Navier-Stokes are:

$$\rho \frac{D\vec{V}}{Dt} = \rho g - \nabla P + \mu \nabla^2 \vec{V}$$

- $\rho \frac{D\vec{V}}{Dt}$: acceleration of the particle
- ρg: gravity particle
- ∇P: pressure forces
- $\mu \nabla^2 \vec{V}$: viscosity forces

Homework chapter 2

1) $\frac{D\rho}{Dt} = 0 \not\Rightarrow \rho = \text{constant}$!

$\frac{D\rho}{Dt} = 0$ means that density of a particle in motion don't change.
not that ρ is constant over the entire flow field

2. the Navier-Stokes equation reduces

$$\begin{cases} 0 = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 u}{\partial y^2}\right) \\ 0 = -\frac{\partial p}{\partial y} - \rho g \\ 0 = -\frac{\partial p}{\partial z} \end{cases} \Rightarrow \begin{cases} \frac{\partial p}{\partial x} = \mu \frac{\partial^2 u}{\partial y^2} \\ p = -\rho g y + f(x) \\ p = f(x, y) \end{cases}$$

In a fully developed flow $\frac{\partial p}{\partial x} = \text{constant}$

then $\begin{cases} \frac{\partial p}{\partial x} = \text{constant} \\ \frac{\partial p}{\partial y} = -\rho g \\ \frac{\partial p}{\partial z} = 0 \end{cases} \Rightarrow \nabla p = \text{constant}$

3 Navier-Stokes equations

$$\rho\left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

$$\rho\left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}\right) = -\rho g - \frac{\partial p}{\partial y} + \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)$$

$x^* = \frac{x}{L}$ $y^* = \frac{y}{L}$ $u^* = \frac{u}{V_\infty}$ $v^* = \frac{v}{V_\infty}$ $p^* = \frac{p}{\rho V_\infty^2}$

$$\Rightarrow \begin{cases} u\frac{\partial u}{\partial x} = V_\infty\left(\frac{u}{V_\infty}\right)\frac{\partial\left(\frac{u}{V_\infty} \times V_\infty\right)}{\partial\left(\frac{x}{L} \times L\right)} = \frac{V_\infty^2}{L} u^* \frac{\partial u^*}{\partial x^*} \\ \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial\left(\frac{y}{L}\right) \times L} \times \left(\frac{\partial\left(\frac{u}{V_\infty}\right) \times V_\infty}{\partial\left(\frac{y}{L}\right) \times L}\right) = \frac{V_\infty}{L^2}\frac{\partial^2 u^*}{\partial y^{*2}} \end{cases}$$

Navier stokes :

$\frac{1}{Re}$ if small Re this ferme become huge

$$u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} = -\frac{\partial p^*}{\partial x^*} + \frac{\mu}{\rho v_\infty L}\left(\frac{\partial^2 u^*}{\partial x^{*2}} + \frac{\partial^2 u^*}{\partial y^{*2}}\right)$$

$$u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} = -\frac{gL}{v_\infty^2} - \frac{\partial p^*}{\partial y^*} + \frac{\mu}{\rho v_\infty L}\left(\frac{\partial^2 v^*}{\partial x^{*2}} + \frac{\partial^2 u^*}{\partial y^{*2}}\right)$$

Large Re: flow behave like an inviscid flow like no viscosity

small Re: viscous forces dominate

Homework chapter 3.

1 the no slip condition produces strong velocity gradients at the wall, so $\omega = \frac{\partial u}{\partial y}$ is create by viscous shear, then the wall is a vorticity source

2. The criterion is $\tau_w = \mu \left.\frac{\partial u}{\partial y}\right|_{y=0} = 0$ at the separation point if flow is two-dimensional and steady τ_w is the shear stress

3 $\frac{\partial u}{\partial t} \sim \nu \frac{\partial^2 u}{\partial y^2}$ diffusion equation

$$\Rightarrow \frac{U_\infty}{t} \sim \nu \frac{u}{\delta^2}$$

$$\Rightarrow \delta \sim \sqrt{\nu t}$$

4. let $\eta = \frac{y}{\delta(x)}$, $\frac{u(x,y)}{U_\infty(x)} = f(\eta)$

these are the conditions

- Steady, incompressible flow
- Simple geometry
- Zero or constant pressure gradient
- High Reynolds number

this way the boundary layer flow be self-similar ?

Homework chapter 4

1) $\delta^* = \int_0^\infty \left(1 - \frac{u}{u_e}\right) dy$ => it's the distance which the external flow is displaced

$\theta = \int_0^\infty \frac{u}{u_e}\left(1 - \frac{u}{u_e}\right) dy$ => measures the loss of momentum in the boundary layer compared to an ideal inviscid flow:

2) $\delta^* = 1{,}7208 \sqrt{\frac{\nu x}{u_\infty}}$, $\theta = 0{,}664 \sqrt{\frac{\nu x}{u_\infty}}$

$$H = \frac{\delta^*}{\theta} = \frac{1{,}7208}{0{,}664} = 2{,}59$$

Stagnation point

$$\delta^* = 1{,}21 \sqrt{\frac{\nu}{a}} \qquad \theta = 0{,}664 \sqrt{\frac{\nu}{a}}$$

$$H = \frac{1{,}21}{0{,}664} = 1{,}82$$

3.) $u_s(\theta) = 2 u_\infty \sin\theta$



$$\frac{du_s(\theta)}{ds} = \frac{2u_\infty}{R}\cos\theta$$

$$\lambda = \frac{\theta^2}{\nu}\frac{dU_\infty}{ds}$$

$$\lambda = \frac{2\theta^2 u_\infty}{\nu R}\cos\theta$$

$$\theta = \sqrt{\frac{R\lambda\nu}{2u_\infty \cos\theta}}$$